Arista Access Node BAS-ACC-54 Site Location Basavanagudi POP Hardware/Environmental Fault Troubleshooting and Resolution SOP

Node: BAS-ACC-54 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_2236

Description

The occurrence of Hardware/Environmental Faults on Arista Access Nodes such as BAS-ACC-54 located at Basavanagudi POP is often attributed to ASIC memory parity errors and line card crashes. This is primarily due to the intricate relationship between ASIC memory parity errors, line card crashes, and power supply redundancy loss. In Arista switches, the ASIC memory parity error is detected by the switch's built-in error detection mechanisms, which trigger an interrupt and cause the switch to generate an alarm. This alarm triggers the Access Node to attempt to recover from the fault by restarting the line card. However, if the power supply redundancy loss occurs, the line card may not be able to recover, resulting in a Critical failure. Furthermore, the power supply redundancy loss can cause the ASIC memory parity error to persist, leading to continuous line card crashes and further exacerbating the failure.

Pre-requisite Checks

- Verify the site location and node ID via 'show version' command on BAS-ACC-54.
- Check the power supply redundancy on BAS-ACC-54 via 'show power supply' command.
- Verify the redundant link status on BAS-ACC-54 via 'show interfaces' command.

Potential Root Causes

Cause	Probability
ASIC memory parity error	High
Line card crash	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show system internal switch fabric fabric-1 fabric-state, Expected Output: Fabric state: Operational, Description: This command will verify if the fabric is operational and check for any fabric-related errors.

Command: show system internal switch fabric fabric-1 fabric-cfg, Expected Output: Fabric configuration: Fabric-1, Description: This command will verify the fabric configuration and check for any fabric-related errors.
```

```
Command: show system internal switch fabric fabric-1 fabric-errors, Expected Output: No fabric errors detected, Description: This command will verify if there are any fabric-related errors.
```

Resolution Steps

Step 1: Step: Replace the faulty ASIC, Command: replace system internal switch fabric fabric-1 ASIC-1, Description: Replace the faulty ASIC with a new one. Verify the new ASIC is operational via 'show system internal switch fabric fabric-1 fabric-state' command.

Step 2: Step: Replace the faulty line card, Command: replace system internal switch fabric fabric-1 line-card-1, Description: Replace the faulty line card with a new one. Verify the new line card is operational via 'show system internal switch fabric fabric-1 fabric-state' command.

Step 3: Step: Replace the power supply, Command: replace power supply 1-1, Description: Replace the faulty power supply with a new one. Verify the new power supply is operational via 'show power supply' command.

Post-Check Verification

```
Command: show system internal switch fabric fabric-1 fabric-state, Expected Output: Fabric state: Operational, Description: Verify the fabric is operational and check for any fabric-related errors.

Command: show system internal switch fabric fabric-1 fabric-cfg, Expected Output: Fabric configuration: Fabric-1, Description: Verify the fabric configuration and check for any fabric-related errors.

Command: show system internal switch fabric fabric-1 fabric-errors, Expected Output: No fabric errors detected, Description: Verify if there are any fabric-related errors.
```

Roll-back Procedures

- Step: Revert the replacement of the ASIC or line card, Command: undo replace system internal switch fabric fabric-1 ASIC-1, Description: Revert the replacement of the ASIC or line card to its original state. Verify the original ASIC or line card is operational via 'show system internal switch fabric fabric-1 fabric-state' command.

- Step: Revert the replacement of the power supply, Command: undo replace power supply 1-1, Description: Revert the replacement of the power supply to its original state. Verify the original power supply is operational via 'show power supply' command.